# PSEUDO-COHERENT COMPLEXES HAVE QUASI-COHERENT COHOMOLOGY IN HIGH DEGREES

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*Editorial note.* Counted span is the author's own proof environment, source lines 3313-3321, immediately following the statement at 3307-3312.

*Editorial note.* The local representative reading is the span's own step; quasi-coherence glues since it is local.

*Allowed standard prerequisites (editorial).*

- Strictly perfect local representatives. definition of m-pseudo-coherence invoked at proof lines 3313-3315. Locally H^i(E) is a cohomology of a strictly perfect complex.

**Lemma 0.1.** *Let $S$ be a scheme. Let $X$ be an algebraic space over $S$. If $E$ is an $m$-pseudo-coherent object of $D(\mathcal{O}_X)$, then $H^i(E)$ is a quasi-coherent $\mathcal{O}_X$-module for $i > m$. If $E$ is pseudo-coherent, then $E$ is an object of $D_{QCoh}(\mathcal{O}_X)$.*

*Proof.* Locally $H^i(E)$ is isomorphic to $H^i(\mathcal{E}^\bullet)$ with $\mathcal{E}^\bullet$ strictly perfect. The sheaves $\mathcal{E}^i$ are direct summands of finite free modules, hence quasi-coherent. The lemma follows. □

*Date*: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.